

# Probabilistic Machine Learning

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## Lecture 06: Evaluation and Empirical Risk Minimisation

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Evaluation units reused verbatim from the data-science course the ERM unit is new and placed last, so it arrives as the theory that explains the practice the students have just been taught rather than as an abstraction preceding it.

# Module 1

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## What Are We Measuring?

**Guiding Question:** *A model reports 99.9% accuracy and has never detected a single case. What should we have measured instead?*

# What We Want to Know

The **true misclassification rate** of a hypothesis  $h$  is the probability that it errs on a randomly drawn object:

$$\text{err}^*(h) = P(h(\mathbf{x}) \neq y(\mathbf{x}))$$

It is defined over the **whole distribution of objects** — which we cannot observe.

## Speaker notes

Deliberately parallel to the population/sample distinction from the statistics module:  $\text{err}^*$  is a population quantity. Students who see the analogy will not be surprised by cross-validation.



# What We Can Measure

We estimate  $\text{err}^*(h)$  by the error on a set of examples  $D_s \subseteq D$  whose true labels are known:

$$\text{err}(h, D_s) = \frac{1}{|D_s|} \sum_{\mathbf{x} \in D_s} 1[h(\mathbf{x}) \neq y(\mathbf{x})]$$

The estimate is only as good as the sample. Which examples land in  $D_s$  — and whether  $h$  has seen them before — decides whether the number means anything.

Everything in the next unit follows from that warning. Ask what happens if  $D_s$  is the data the model was fitted on the answer motivates hold-out estimation without needing to be told.



# A Number Alone Means Nothing

"95% accuracy" is not a result until you know what a trivial method achieves.

## Trivial classification

Always predict the **majority class**.

## Trivial regression

Always predict a **central tendency** — the mean or median of  $y$ .

## Random guessing

Draw a label from the marginal distribution  $p(y)$ .

Insist on the habit: report the baseline next to the model, always. This is the cheapest defence against self-deception in the whole course.



# Example: Why Accuracy Deceives

A screening test for a disease affecting **1 in 1,000** people.

## The "model"

Always predict *healthy*.

## Its accuracy

**99.9%** — and it has never detected a single case.

On imbalanced data the majority baseline is nearly unbeatable by accuracy, and completely useless. The metric, not the model, is the problem.

The same base-rate structure as the Bayes unit, now as an evaluation trap. This is the motivation for precision and recall, which follow immediately.

# Module 2

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## Estimating Error Honestly

**Guiding Question:** *Every error number comes from a sample. Which sampling schemes give a number you may quote?*

# Training Error Is Not an Estimate

When the evaluation set equals the training set,  $D_{tr} = D$ , the resulting number measures how well the model **memorised**, not how well it will generalise.

Training error is a lower bound on the true error, and an arbitrarily bad one. A model with enough capacity can drive it to zero while learning nothing.

Connect straight to the bias–variance unit: training error falls monotonically with capacity, which is exactly why it cannot be the criterion for choosing capacity.

# Four Estimation Schemes

## Hold-out (2-fold)

Split  $D$  into a training and a test part,  $D_{tr} \subset D$ . Fit on one, evaluate on the other. *Simple, but wastes data and depends on the split.*

## $k$ -fold cross-validation

Partition  $D$  into  $k$  folds; each fold serves once as the test set. Average the  $k$  errors. *Uses all data for both roles  $k = 5$  or  $10$  is customary.*

## Leave-one-out ( $n$ -fold)

$k = n$ : every example is a test set of size one. *Nearly unbiased, expensive, and high variance.*

## Bootstrapping

Resample  $D$  with replacement to build training sets evaluate on the examples left out. *Useful when  $D$  is small.*

Ask which scheme a Kaggle-style leaderboard implements, and what repeatedly consulting a fixed test set does to its status as an estimate — the test set becomes training data for the modeller.

# Misclassification Costs

Not every error costs the same. A missed diagnosis and a false alarm have different consequences, so counting errors equally is itself a modelling assumption.

A **cost matrix** assigns a cost to each (true class, predicted class) pair; the objective becomes expected cost rather than error rate.

Speaker notes

This is the hinge into precision and recall in two units' time. Ask the group to set the two costs for a spam filter and then for a cancer screen the ratios they choose differ by orders of magnitude, and both are defensible.



# Takeaway

An error number is meaningless without the sampling scheme that produced it. **Never evaluate on data the model was fitted on**, and state the cost model when errors are not interchangeable.

# Module 3

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## Which Kind of Wrong?

**Guiding Question:** *Being imprecise and being incomplete are different failures. Which one does your application care about?*

# Why the Error Rate Is Not Enough

For multiclass classification  $h_{\Theta} : X \rightarrow Y$  with  $Y \subseteq \{1, \dots, N\}$ , the plain error rate:

## Fails on imbalance

The majority baseline scores high while detecting nothing.

## Hides the distribution

Which classes the errors fall on stays invisible.

## Hides the kind of error

Missing assignments and imprecise assignments count the same.

Three distinct failures, deliberately separated: imbalance, distribution across classes, and type of error. Each one motivates a different part of what follows.

# The Confusion Matrix

Rows are true classes, columns are predictions cell  $(i, j)$  counts the examples of class  $i$  that  $h$  assigned to  $j$ . The diagonal is correct, and every off-diagonal cell names a specific confusion.

The matrix is the raw material: every measure below is a ratio of its cells. Read it before reading any summary number.

On a well-known dataset the off-diagonal structure is often interpretable — which two classes get confused tells you something about the representation, not just the score.

# Four Counts, Per Class

For a class  $j$ , every prediction falls into one of four cells:

$TP_j$  — **true positives**

Assigned to  $j$  **and** truly in  $j$ .

$FP_j$  — **false positives**

Assigned to  $j$  but **not** in  $j$ .

$FN_j$  — **false negatives**

**Not** assigned to  $j$  but truly in  $j$ .

$TN_j$  — **true negatives**

Not assigned to  $j$  and not in  $j$ .

Read the letters as **decision then correctness**: P/N is what  $h$  decided, T/F is whether that decision was right.

The mnemonic in the tip box removes nearly all of the confusion students have with these four terms. Have them classify a concrete example aloud before moving on.

# Precision and Recall

## Recall — completeness

$$\text{rec}_j = \frac{TP_j}{TP_j + FN_j}$$

Of everything that truly belongs to  $j$ , what fraction did  $h$  find?

## Precision — exactness

$$\text{prec}_j = \frac{TP_j}{TP_j + FP_j}$$

When  $h$  says  $j$ , how often is it right?

Both are **conditional probabilities with different conditions** — recall conditions on the truth, precision on the prediction. Swapping them is the same error as swapping the conditionals in Bayes' theorem.

Speaker notes

The link to Bayes is the payoff of teaching probability first: precision and recall are  $P(\text{correct} \mid \text{predicted})$  and  $P(\text{found} \mid \text{true})$ . Ask which one a search engine optimises and which one a screening programme optimises.



# Mini-Exercise

A spam filter processes 1,000 emails, 100 of which are spam. It flags 80 emails, of which 60 are genuinely spam.

Compute precision and recall. Which one would you rather improve, and why?

# Mini-Exercise — Solution

The final sentence is the one to stabilise — the choice between precision and recall is a statement about costs, which is why the cost-matrix slide came earlier.

# Mini-Exercise — Solution

$$\text{Precision} = 60/80 = 0.75$$

A quarter of flagged mail is legitimate.

$$\text{Recall} = 60/100 = 0.60$$

Forty spam messages still arrive.

For a spam filter, **precision** usually matters more: a lost legitimate email costs far more than a spam message that slips through. The cost matrix decides, not the metric.

# Takeaway

The confusion matrix shows **which** errors occur precision and recall split "being wrong" into **being imprecise** and **being incomplete**. Which one you optimise is a decision about costs, not about mathematics.

# Squared Error

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (h(\mathbf{x}_i) - y_i)^2 \quad \text{RMSE} = \sqrt{\text{MSE}}$$

## MSE

Average quadratic loss always non-negative, lower is better.

Unit is the **square** of the target's unit — hard to interpret directly.

Large errors are penalised disproportionately.

## RMSE

Same ordering as MSE, but back in the **target's own unit** — which is why it is the one usually reported.

## Speaker notes

The quadratic penalty is a modelling choice with consequences: it makes the fit sensitive to outliers, which connects back to robustness in descriptive statistics. Ask when that sensitivity is desirable — when large errors really are disproportionately costly.



# Absolute and Relative Error

## MAE

$$\frac{1}{n} \sum_i |h(\mathbf{x}_i) - y_i|$$

Easier to interpret than RMSE  
treats all errors proportionally.

## Mean error

Drop the absolute value to  
keep the **direction** — reveals  
systematic over- or under-  
prediction.

## Mean relative error

$$\frac{1}{n} \sum_i \frac{|h(\mathbf{x}_i) - y_i|}{|y_i|}$$

The average error *per unit of target* — comparable across  
scales.

Normalised RMSE variants aim to make errors comparable between datasets, but are hard to interpret. Prefer a measure whose unit you can name.

Speaker notes

The mean error (with sign) is underused and worth highlighting: a model can have excellent RMSE and still be biased in one direction, which matters for forecasting and pricing.



# Takeaway

Squared measures punish large errors and lose the unit absolute measures stay interpretable signed measures expose bias. Report the one whose **unit and failure mode** match the decision the prediction feeds.

# Module 4

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## Empirical Risk Minimisation

**Guiding Question:** *Every method in this course minimises something on the training data. What is the quantity we actually wanted — and what does the substitution cost?*

# What We Actually Want to Minimise

The **risk** of a hypothesis  $h$  under loss  $L$  and the true data distribution  $p(\mathbf{x}, y)$ :

$$R(h) = \mathbb{E}_{(\mathbf{x}, y) \sim p} [L(h(\mathbf{x}), y)]$$

This is the quantity that matters — expected loss on data we have not seen. It is also **uncomputable**:  $p$  is unknown.

Math first (P5). Everything in the lecture follows from the gap between what we want (an expectation under  $p$ ) and what we can measure (an average over a sample). Students have met this as population versus sample; here it is named.

# What We Can Compute

The **empirical risk** — the average loss on the training sample:  $\hat{R}(h) = \frac{1}{n} \sum_{i=1}^n L(h(\mathbf{x}_i), y_i)$  **Empirical risk minimisation (ERM)** chooses  $\hat{h} = \arg\min_{h \in \mathcal{H}} \hat{R}(h)$ .

Nearly every method in this course is ERM with a particular loss and hypothesis space — including maximum likelihood, which is ERM with the **negative log-likelihood** as the loss.

## Squared loss

linear regression

## Cross-entropy

logistic regression, neural networks

## 0/1 loss

what we usually care about, and rarely optimise — it has no useful gradient

The unification is the payoff of the lecture: one principle, many algorithms. The 0/1 remark explains why surrogate losses exist at all — a recurring theme from the perceptron onwards.

# The Generalisation Gap

$$R(h) = \underbrace{\hat{R}(h)}_{\text{measurable}} + \underbrace{\big(R(h) - \hat{R}(h)\big)}_{\text{generalisation gap}}$$

## Why the gap opens

$\hat{h}$  was **chosen using the same data** that then measures it. The minimum of a noisy objective is optimistically biased.

## What controls it

The **capacity** of  $\mathcal{H}$  and the sample size  $n$ .  
Richer hypothesis spaces fit the noise better and generalise worse for a given  $n$ .

This is the bias–variance trade-off stated in terms of risk — and the reason training error is not an estimate of anything.

# Structural Risk Minimisation

Rather than minimising empirical risk alone, minimise it **plus a complexity penalty**:

$$\hat{h} = \arg \min_{h \in \mathcal{H}} \hat{R}(h) + \lambda \Omega(h)$$

## $\Omega$ as capacity control

Restrict effective capacity so the gap stays small.

## $\Omega$ as a prior

The MAP reading:  $\lambda \Omega(h) = -\log p(h)$ .

## Same equation

Regularisation, priors and capacity control are three descriptions of one term.

# Mini-Exercise

A model has training error 0.02 and test error 0.19.

1. Which quantity is  $R(h)$  better estimated by?
2. Why is the training error not simply "a bit optimistic"?
3. Give two ways to shrink the gap, one changing  $\mathcal{H}$  and one not.

# Mini-Exercise — Solution

We want **risk** and can only measure **empirical risk**. ERM minimises the proxy the **generalisation gap** is the price, and every regulariser — read as a penalty or as a prior — is an attempt to control it.

# Mini-Exercise — Solution

1. The **test** error, 0.19 — measured on data not used to choose  $h$ .

2. Because  $h$  was selected to **minimise** the training error. The selection itself introduces the bias the more hypotheses were compared, the larger it is.

3. Change  $\mathcal{H}$ : regularise, or use a smaller model. Leave  $\mathcal{H}$  : collect more data.

We want **risk** and can only measure **empirical risk**. ERM minimises the proxy the **generalisation gap** is the price, and every regulariser — read as a penalty or as a prior — is an attempt to control it.